

Linux command line tutorial

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Overview

- What is Linux & the Terminal
- Navigating the File System
- Files & Directories
- Viewing & Editing Text Files
- Searching & Redirection
- Permissions & Processes
- Using Cerberus Cluster

About Linux

- What is Linux?
 - **Linux** is a **free, open-source operating system** that manages a computer's hardware and lets software run on it. The Linux kernel is the core program that controls the CPU, memory, disks, and networking.
 - In practice, people use **Linux distributions (distros)**, which bundle: the Linux kernel, system tools (GNU utilities, system libraries), a package manager, optional graphical desktop environments. Examples distros: **Ubuntu, Debian, Red Hat Enterprise Linux, Fedora, SUSE, Arch.**
- Market share in the enterprise and general server OS market
 - Linux is the **leading platform** with **60% market share** of all server operating systems
 - **~96% of the top 1 million web servers** run Linux.
 - **85%** of cloud virtual machines, containers, and Kubernetes clusters run Linux
 - Major cloud providers (AWS, Google Cloud, Azure) heavily favor Linux for most workloads.
 - **100 % of the world's top 500 supercomputers** run Linux distributions, making it de facto standard for HPC.

Linux Terminal

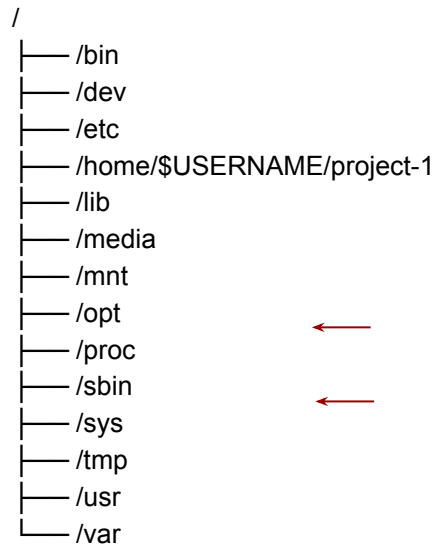
- Need for command line skills
- Terminal vs GUI
- Command structure
 - `$ {command} {options} {arguments}`
 - `$ ls -ltr /home/$USERNAME/`
- Commands are case sensitive

Navigating the File System

- Concepts
 - Directory tree
 - Absolute vs relative paths
- Commands
 - `$ whoami`
 - `$ pwd`
 - `$ hostname`
 - `$ ls`
 - `$ ls -l`
 - `$ ls -a`
 - `$ cd`
 - `$ cd ..`
 - `$ cd ~`

Directory tree

```
/
├── /bin
├── /dev
├── /etc
├── /home/$USERNAME/project-1
├── /lib
├── /media
├── /mnt
├── /opt
├── /proc
├── /sbin
├── /sys
├── /tmp
├── /usr
└── /var
```



Working with Files & Directories

- Concepts
 - Create, move, copy, delete files
 - Tab completion
 - Commands

```
$ pwd
$ mkdir ~/project-1
$ mkdir ~/project-1/project-linux
$ cd ~/project-1/project-linux
$ pwd
$ touch file.txt
$ ls -ltr
$ cp file.txt file_copy.txt
$ ls -ltr
```

```
$ mv file_copy.txt /tmp/
$ rm /tmp/file_copy.txt
$ cd ~
$ rm -r ~/project-1
```

Caution

- rm is permanent
- no recycle bin

Viewing & Editing Text Files

- Concepts
 - Text files are central to Linux & bioinformatics
 - Commands

```
$ cd ~/project-1
$ wget https://data.glygen.org/ln2downloads/tmp/toy.fastq
$ ls -ltr
$ cat toy.fastq
$ less toy.fastq
$ head -20 toy.fastq
$ tail -20 toy.fastq
$ ls -ltr
$ nano file.txt
```

Type the following in file.txt

```
Hello world!
Today, I am starting to study mouse genes.
Isn't that cool?
Genes are ...? I don't know, I am still learning.
I don't want make a mistake.
Just want to say hello.
```

Searching, Pipes & Redirection

- Concepts
 - Everything is text
 - Output can be reused
 - Commands

```
$ cd ~/project-1
$ grep "mistake" file.txt
$ grep -i "gene" file.txt
$ echo "hello" > file.txt
$ echo "world" >> file.txt
$ cat file.txt | grep "hello"
$ ls | grep ".txt"
```


Permissions & Processes

- File permissions
 - r = read, w = write, x = execute
 - Commands

File permissions

```
$ cd ~/project-1  
$ ls -l  
$ chmod 755 file.txt
```

Processes

```
$ ps  
$ top  
$ htop  
  
Ctrl + C
```

Wrap-Up & Best Practices

- Key habits
 - Use Tab for autocomplete
 - Use ↑ for command history
 - Use man pages
 - ls --help

Homework / Practice

- Create 3 text files
- Write text into them
- Search for a word using grep
- Delete one file safely

Bioinformatics Project Layout

- Concepts
 - Absolute vs relative paths
 - Standard project structure
- Key points
 - Never mix raw data and results
 - Reproducibility starts with organization
- Commands

```
$ cd ~/
$ mkdir -p ~/project-rnaseq/data
$ mkdir -p ~/project-rnaseq/results
$ mkdir -p ~/project-rnaseq/scripts
$ mkdir -p ~/project-rnaseq/logs
$ cd ~/project-rnaseq/
$ ls -ltr
```

```
├─ project-rnaseq
│  ├── data
│  ├── logs
│  ├── results
│  └── scripts
```

Managing Sequencing Files

- Concepts
 - FASTQ files
 - Compression (.gz)
 - File sizes
- Key points
 - Why files are compressed
 - Why you don't unzip large FASTQs
- Commands

```
$ cd ~/project-rnaseq/  
$ cp ~/project-1/toy.fastq ./data/  
$ ls -lh ./data  
$ du -sh ./data
```

```
$ gzip -c ./data/toy.fastq > ./data/toy.fastq.gz  
$ ls -lh ./data
```

Inspecting Bioinformatics Files

- Linux tools are used to sanity-check data before analysis
- Structure of FASTQ and GTF files
- Commands

```
$ cd ~/project-rnaseq/  
$ less ./data/toy.fastq  
$ zcat ./data/toy.fastq.gz  
$ zcat ./data/toy.fastq.gz | grep "^@" | head  
$ zcat ./data/toy.fastq.gz | wc -l
```

```
$ cd ~/project-rnaseq/  
$ head -20 ./data/toy.fastq  
$ grep "^1" ./data/toy.fastq  
$ grep -v "^#" ./data/toy.fastq
```

Best Practices & What Comes Next

- Essential habits
 - Never analyze in /data
 - Use scripts, not manual commands
 - Document everything